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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree Supplementary Examination October 2023 (2019 Scheme)

Course Code: EET424

Course Name: ENERGY MANAGEMENT

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|---|-----|
| 1 | Explain the objectives of energy management. | (3) |
| 2 | Write short notes on Building Management Systems. | (3) |
| 3 | Discuss how energy efficient motors help in reducing energy consumption. | (3) |
| 4 | How motor load matching is helpful in improving the efficiency of motor | (3) |
| 5 | What is meant by time-of-day pricing? | (3) |
| 6 | What do you mean by strategic conservation. How it is helpful in demand side management | (3) |
| 7 | Explain two different types of co-generation systems | (3) |
| 8 | Discuss the difference between COP and efficiency in the case of a refrigerator. | (3) |
| 9 | What are the advantages and disadvantages of the payback period method? | (3) |
| 10 | Explain the advantages of computer aided energy management system. | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

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|----|---|------|
| 11 | a) Explain different steps involved in energy management planning | (10) |
| | b) What is energy audit report? Explain. | (4) |

OR

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|----|--|------|
| 12 | a) Why power quality auditing is to be carried out? Will it result in improving the energy efficiency of a plant? Explain. | (10) |
| | b) Write short notes on Energy Conservation Building Code. | (4) |

Module II

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|----|--|-----|
| 13 | a) Explain any six methods to reduce energy consumption in lighting. | (8) |
| | b) Explain cascade efficiency in an electrical system. Discuss how it can be improved. | (6) |

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- 14 a) Explain the methods to reduce energy consumption in motors. (10)
b) Write notes on design measures for increasing efficiency in transformers. (4)

Module III

- 15 a) Explain the benefits of power factor improvement. Derive an expression for the most economical value of power factor which may be attained by a consumer. (7)
b) Discuss the importance of peak demand control. Explain the different peak demand control methods in detail. (7)

OR

- 16 a) What is meant by ancillary services? Explain the different types of ancillary services. (7)
b) The load on an installation is 800 kW, 0.8 lagging p.f. which works for 3000 hours per annum. The tariff is Rs 100 per kVA plus 20 paise per kWh. If the power factor is improved to 0.9 lagging by means of loss-free capacitors costing Rs 60 per kVAR, calculate the annual saving effected. Allow 10% per annum for interest and depreciation on capacitors. (7)

Module IV

- 17 a) Explain any five opportunities for energy savings in steam distribution. (7)
b) Explain any four energy conservation opportunities in furnace. (7)

OR

- 18 a) What are the needs of boiler blowdown? Explain different blowdown methods. (7)
b) Explain any five energy saving opportunities in heating, ventilating and air conditioning systems. (7)

Module V

- 19 a) What is meant by life cycle costing method? Explain the steps in life cycle costing approach for the selection of energy projects. (6)
b) Calculate the energy saving and payback period which can be achieved by replacing a 11 kW, existing motor with an EEM. The capital investment required for EEM is Rs. 40,000/-. Cost of energy/kWh is Rs. 5. The loading is 70% of the rated value for both motors. Efficiency of the existing motor is 81% and that of EEM is 84.7%. (8)

OR

- 20 a) Explain net present value method for selection of energy projects. Mention its advantages and drawbacks. (6)
b) Calculate the internal rate of return (IRR) for a device that costs 5 lakhs which lasts for 10 years and result in fuel saving of Rs 1.5 lakhs each year. (8)
